Lecture 17 2025-10-28

Forward differences (special case of equidistant interpolation points)

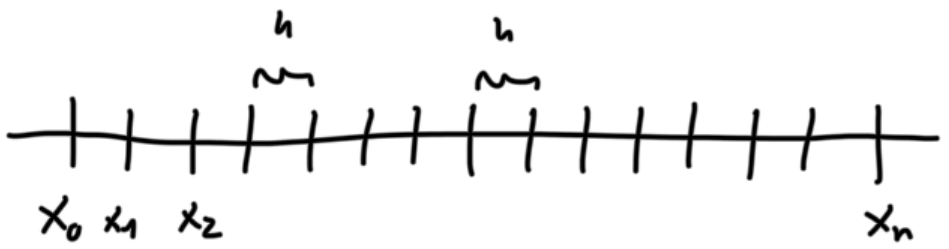
$x_0 = 0$

$x_j = jh$ (j integer) $\quad x = sh \quad h$ = grid spacing / mesh size

s: "dimensionless x" ($s_0 = 0,\ s_1 = 1,\ s_2 = 2$ etc.)

$x_0 \neq 0$: $\quad x_j = x_0 + jh$

$x = x_0 + sh$

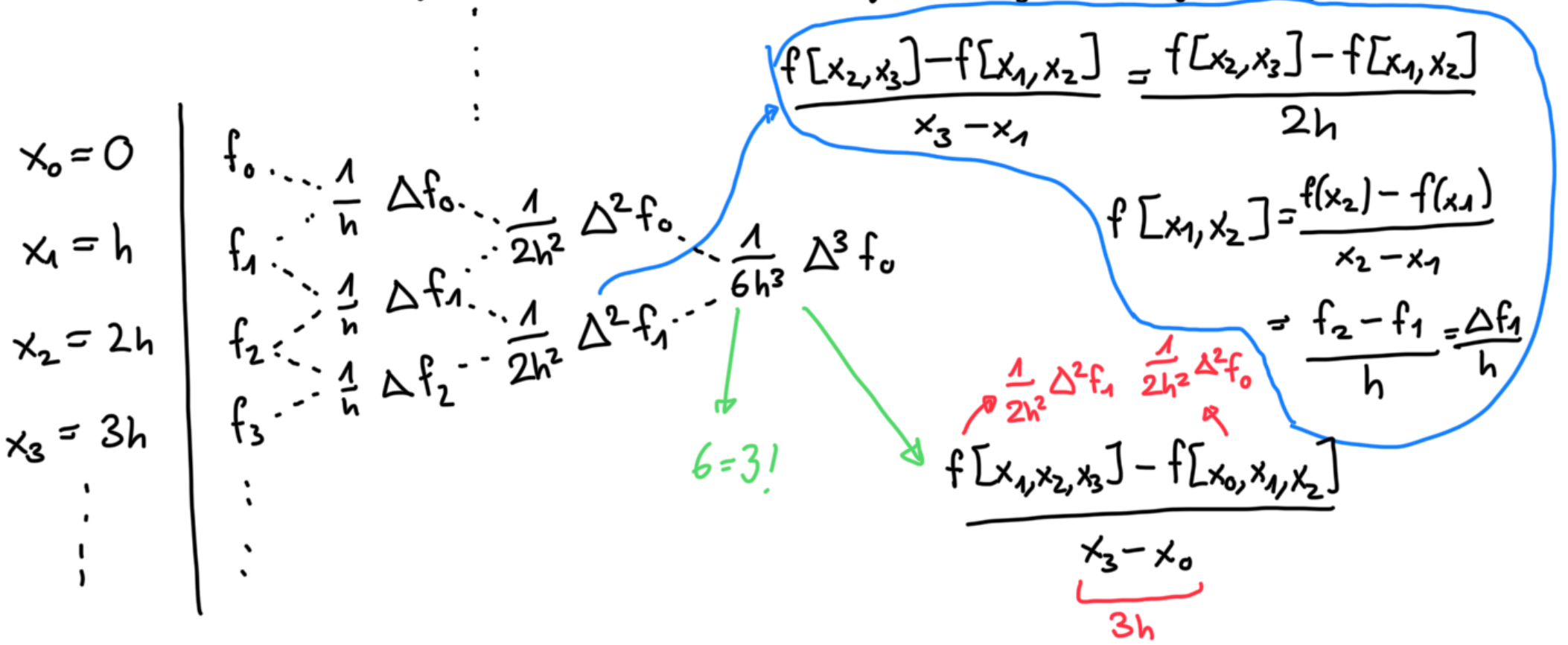


Notation: $f_j = f(x_j)$, $\Delta f_j = f_{j+1} - f_j$

$$\Delta^2 f_j = \Delta f_{j+1} - \Delta f_j = f_{j+2} - 2f_{j+1} + f_j$$

$$\Delta^3 f_j = \Delta^2 f_{j+1} - \Delta^2 f_j = f_{j+3} - 3f_{j+2} + 3f_{j+1} - f_j$$

$$\vdots$$



General formula: $f[x_0, \dots, x_k] = \frac{1}{k!\,h^k}\,\Delta^k f_0$

$$P_{0:n}(x) = f[x_0] + \sum_{k=1}^{n} f[x_0,\dots,x_k] \underbrace{\prod_{j=0}^{k-1} (x - x_j)}_{(*)}$$

When the points are uniformly spaced

$$(*) = \prod_{j=0}^{k-1} (s-j)h = h^k s(s-1)\cdots(s-k+1) = h^k\, k! \binom{s}{k}$$

$$\text{where } \binom{s}{k} = \frac{s(s-1)\cdots(s-k+1)}{k!}$$

Plugging this into the formula for $P_{0:n}(x)$, we get

$$P(x) = f_0 + \sum_{k=1}^{n} \frac{1}{k!h^k} \Delta^k f_0\, h^k\, k! \binom{s}{k} = f_0 + \sum_{k=1}^{n} \binom{s}{k} \Delta^k f_0$$

$$(x = sh)$$

Remark: One can do the same with backward differences

$\nabla f_j := f_j - f_{j-1}$ as well (see textbook p. 127–128, (3.13))

Theorem 3.6 If $f \in C^n([a,b])$ and $x_0, \dots x_n \in [a,b]$ are distinct then there exists $\xi(x) \in (a,b)$ such that

$$f[x_0, \dots x_n] = \frac{f^{(n)}(\xi(x))}{n!}.$$

Proof: $g(x) = f(x) - P_{0:n}(x)$

We have $g(x_j) = f(x_j) - P_{0:n}(x_j) = 0$ for $j = 0, \dots, n$

since $P_{0:n}(x)$ interpolates f at x_j

$\Rightarrow$ g has (at least) $n+1$ zeros

Rolle's theorem $\Rightarrow$ g' has (at least) n zeros

$\vdots$

$g^{(n)}(x)$ has at least one zero, call it $\xi(x)$

(generalized Rolle's theorem)

$$0 = g^{(n)}(\xi(x)) = f^{(n)}(\xi(x)) - \frac{\partial^n}{\partial x^n} P_{0:n}(\xi(x))$$

So $$f^{(n)}(\xi(x)) = \frac{\partial^n}{\partial x^n} P_{0:n}(\xi(x)) = \frac{d^n}{dx^n}(a_n x^n) = n!\, a_n$$

where $a_n = f[x_0, \dots, x_n]$ = leading coefficient of $P_{0:n}(x)$

(see previous theorem)

$$\Rightarrow \frac{1}{n!} f^{(n)}(\xi(x)) = f[x_0, \dots, x_n]$$ ∎

3.4 Hermite Interpolation

Generalization of Lagrange and Taylor polynomials

We are again given $n+1$ nodes $x_0, \dots, x_n$

Now we give each node a multiplicity $m_i + 1$:

$m_i = 0 \rightarrow$ usual case

$m_i = 1 \rightarrow$ double node

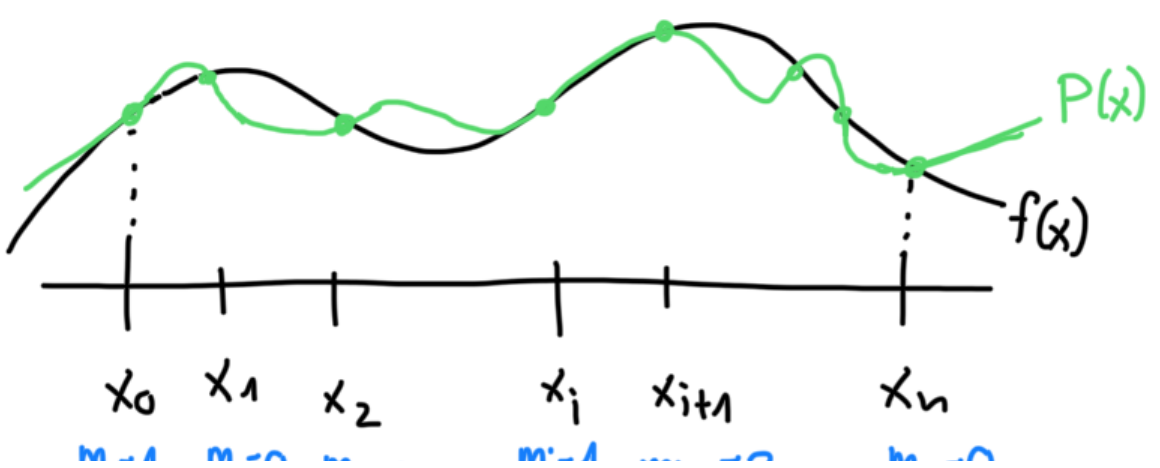
$m_i = 2 \rightarrow$ triple node

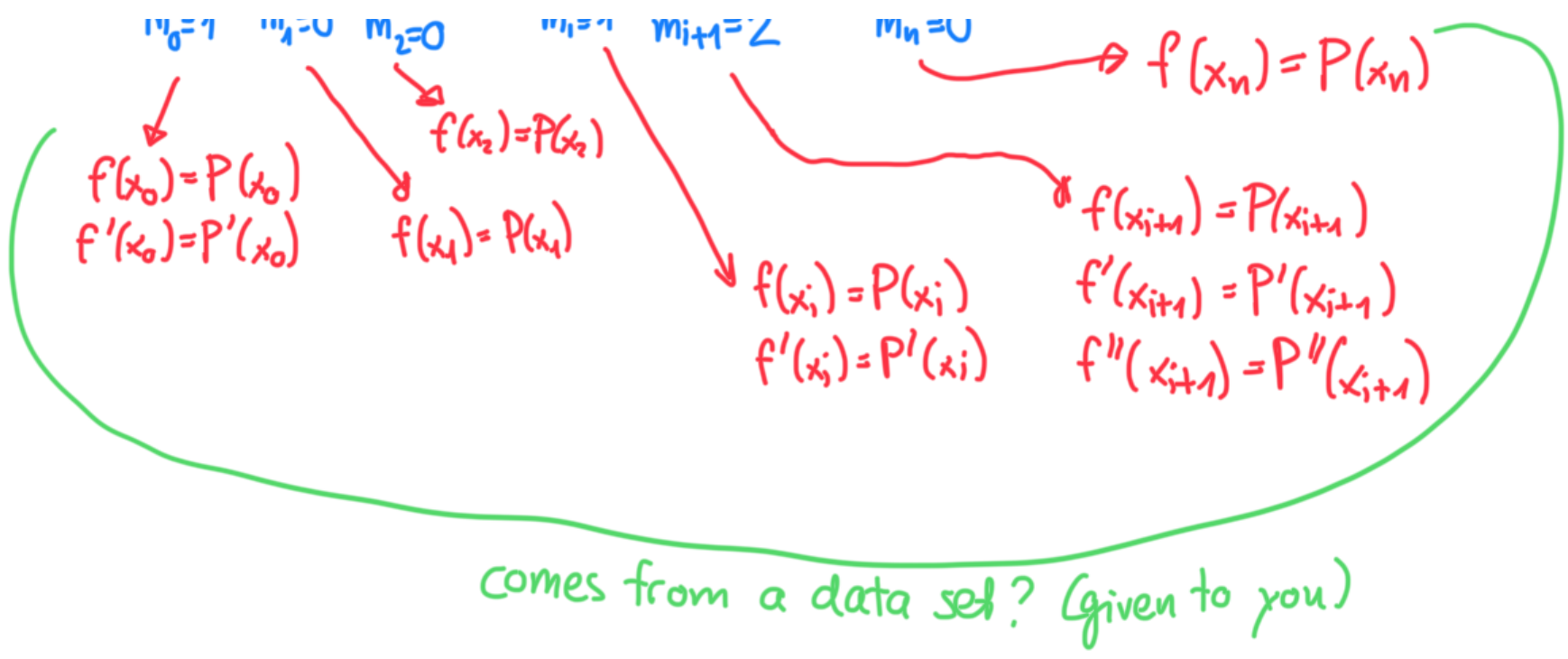
etc.

$$\deg P(x) \leq n + \sum_{i=0}^{n} m_i = \begin{cases} n & \text{if all } m_i = 0 \\ 2n+1 & \text{if all } m_i = 1 \end{cases}$$

Matching conditions: $P^{(k)}(x_i) = f^{(k)}(x_i)$ $\quad 0 \leq i \leq n$, $0 \leq k \leq m_i$

Total number of matching conditions: $n+1 + \sum_{i=0}^{n} m_i$





This is called an <u>osculating polynomial</u> approximating f

<u>Hermite Interpolation</u>: Special case when all the m_i are equal to 1.

Basis functions (play same role as Lagrange interpolating polynomials)

$$H(x) = H_{0:n}(x) = \sum_{j=0}^{n} f(x_j) H_j(x) + \sum_{j=0}^{n} f'(x_j) \hat{H}_j(x)$$

H, H_j, $\hat{H}_j$ are polynomials of degree $2n+1$

with (6)
$$\begin{cases} H_k(x_j) = \delta_{jk} \;, \quad \hat{H}_k(x_j) = 0 \\ H_k'(x_j) = 0 \;, \quad \hat{H}_k'(x_j) = \delta_{jk} \end{cases} \qquad j = 0, \dots, n \;, \; k = 0, \dots, n$$

$$\delta_{jk} = \begin{cases} 1 & \text{if } j=k \\ 0 & \text{otherwise} \end{cases}$$

where $H_k(x) = (1 - 2(x-x_k) L_k'(x_k))(L_k(x))^2$

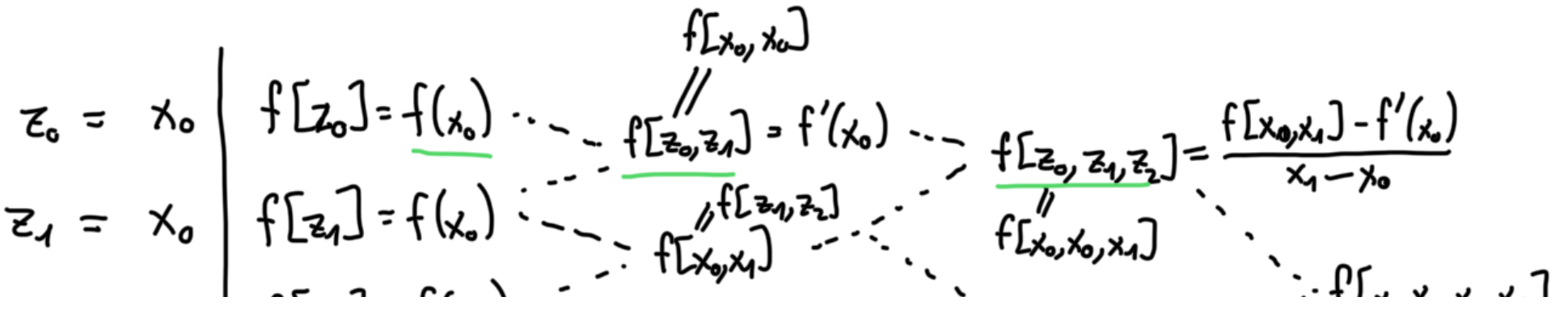
$$\hat{H}_k(x) = (x-x_k)(L_k(x))^2$$

<u>Exercise</u>: Check that these satisfy (6) (Sec 3.4 Exercise 11a)

<u>Textbook</u>: $H_{2n+1}(x)$ for $H(x)$

$H_{n,k}$ and $\hat{H}_{n,k}$ for H_k & $\hat{H}_k$ basis functions

We can use divided differences to form H (computing all the H_k and $\hat{H}_k$ is a pain)



$z_2 = x_1$	$f[z_2] = f(x_1)$				
$z_3 = x_1$	$f[z_3] = f(x_1)$	$f[x_1, x_1] = f'(x_1)$		$f[x_0, x_1, x_1]$	
$z_4 = x_2$	$f[z_4] = f(x_2)$	$f[x_1, x_2]$		$f[x_1, x_1, x_2]$	
$z_5 = x_2$	$f[z_5] = f(x_2)$	$f[x_2, x_2] = f'(x_2)$			

$f[x_0, x_0, x_1, x_1]$ = ... = $\dfrac{f[x_0, x_1, x_1] - f[x_0, x_0, x_1]}{x_1 - x}$

Heuristics:

x_0
$x_0 + \varepsilon$
x_1
$x_1 + \varepsilon$
⋮

take $\varepsilon \to 0$ to obtain this table

then $f[x_0, x_0+\varepsilon] = \dfrac{f(x_0+\varepsilon) - f(x_0)}{\varepsilon} \to f'(x_0)$ as $\varepsilon \to 0$

$f[x_1, x_1+\varepsilon] = \dfrac{f(x_1+\varepsilon) - f(x_1)}{\varepsilon} \to f'(x_1)$ as $\varepsilon \to 0$

$f[x_0, x_0] = f[z_0, z_1]$

So we obtain $H(x) = f(x_0) + f'(x_0)(x - \underbrace{x_0}_{z_0}) + f[x_0, x_0, x_1](x-x_0)^2 + f[x_0, x_0, x_1, x_1](x-x_0)^2(x-x_1) + f[x_0, x_0, x_1, x_1, x_2](x-x_0)^2(x-x_1)^2 + \dots$

$(x-z_0)(x-z_1)$; $(x-z_0)(x-z_1)(x-z_2)$; $(x-z_0)(x-z_1)(x-z_2)(x-z_3)$

$z_0\ z_1\ z_2\ z_3$; $z_0\ z_1\ z_2\ z_3\ z_4$

Lecture 18 2025-10-30

Theorem 3.9 If $f \in C^{2n+2}([a,b])$ then

$$f(x) = H_{0:n}(x) + \frac{f^{(2n+2)}(\xi(x))}{(2n+2)!}(x-x_0)^2 \dots (x-x_n)^2$$

for some $\xi(x) \in [a,b]$

Proof: Exercise 11 Section 3.4

Example: Let $f(x) = \sin \pi x$. Find $P(x)$ of degree 3 s.t.

$P(0) = f(0)$
$P(\frac{1}{2}) = f(\frac{1}{2})$
$P'(\frac{1}{2}) = f'(\frac{1}{2})$
$P''(\frac{1}{2}) = f''(\frac{1}{2})$

$f'(x) = \pi \cos(\pi x)$

$f''(x) = -\pi^2 \sin(\pi x)$

Use $f[\underbrace{x_0, \dots, x_0}_{n+1 \text{ times}}] = \dfrac{f^{(n)}(x_0)}{n!}$ (*)

$\dfrac{f[x_1, x_1] - f[x_0, x_1]}{x_1 - x_0}$

$n+1$ times

Solution: Construct table with divided differences

$z_0 = x_0 = 0$

$z_1 = x_1 = \frac{1}{2}$

$z_2 = x_1 = \frac{1}{2}$

$z_3 = x_1 = \frac{1}{2}$

$f[x_i] = f(x_i)$	$f[z_i, z_{i+1}]$	$f[z_i, z_{i+1}, z_{i+2}]$	
0	$f[x_0,x_1] = \frac{1-0}{\frac{1}{2}-0} = 2$	$\frac{0-2}{\frac{1}{2}-0} = -4$	$\frac{-\frac{\pi^2}{2}-(-4)}{\frac{1}{2}-0} = 8-\pi^2$
1	$f[x_1,x_1] = f'(x_1) = 0$ (1)	$f[x_1,x_1,x_1] = \frac{f''(x_1)}{2} = -\frac{\pi^2}{2}$	
1	$f[x_1,x_1] = f'(x_1) = 0$ (2)		
1			

$f(x_1)-f(x_0)$ / x_1-x_0

$$P(x) = 0 + 2(x-0) - 4(x-0)(x-\tfrac{1}{2}) + (8-\pi^2)(x-0)(x-\tfrac{1}{2})^2$$

($x_0 = 0$, $x_0 = 0$, $x_1 = \frac{1}{2}$, $x_0 = 0$, $x_1 = \frac{1}{2}$)

Estimate error at $x = \frac{1}{4}$: (Thm 3.9)

$$R(x) = \frac{f^{(4)}(\xi)}{4!} x(x-\tfrac{1}{2})^3$$

$$f^{(4)}(x) = \pi^4 \sin(\pi x) \Rightarrow |f^{(4)}(\xi)| \leq \pi^4 \quad \text{for } \xi \in [0,1]$$

$$|R(\tfrac{1}{4})| \leq \frac{\pi^4}{4!}\left(\frac{1}{4}\right)^4 = \dots$$

Chapter 4: Numerical Differentiation and Integration

4.1. Numerical Differentiation

Recall Definition of Derivative: $f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$

$\Rightarrow$ For small h, we expect $f'(x_0) \approx \frac{f(x_0+h)-f(x_0)}{h}$

Questions: • What is the error?

Questions: · What is the error?

- More generally, suppose we know $f(x)$ at $x_0, \dots, x_n$. What linear combination of these values best approximates $f'(x)$ at $x_0, \dots, x_n$?
- Higher order derivatives?

Idea: Lagrange polynomial $P_n(x) = \sum_{k=0}^{n} f(x_k) L_k(x)$ is a good approximation of f $\rightsquigarrow$ maybe its derivative is a good approximation of f'?

We know $f \in C^{n+1}([a,b])$, we have

$$f(x) = \sum_{k=0}^{n} f(x_k) L_k(x) + u(x) \prod_{k=0}^{n} (x - x_k) \qquad (\circledcirc)$$

where $u(x) = \dfrac{f^{(n+1)}(\xi(x))}{(n+1)!}$ for $\xi \in [x_0, x_n]$

(or $[x_n, x_0]$ if $x_n < x_0$)

$$u(x) = \frac{f(x) - \sum_{k=0}^{n} f(x_k) L_k(x)}{\prod_{k=0}^{n} (x - x_k)} = f[x_0, \dots, x_n, x]$$

So if we take the derivative of $(\circledcirc)$ we get

$$f'(x) = \sum_{k=0}^{n} f(x_k) L_k'(x) + u'(x) \prod_{k=0}^{n} (x - x_k) + u(x) \sum_{\ell=0}^{n} \prod_{\substack{k=0 \\ k \neq \ell}}^{n} (x - x_k)$$

$u'(x)$: hard to evaluate

$\prod_{k=0}^{n} (x - x_k)$: $= 0$ at x_j

$\sum_{\ell=0}^{n} \prod_{\substack{k=0 \\ k \neq \ell}}^{n} (x - x_k)$: gives error estimate; all terms drop except $j = \ell$ if $x = x_j$

Plug in $x = x_j$

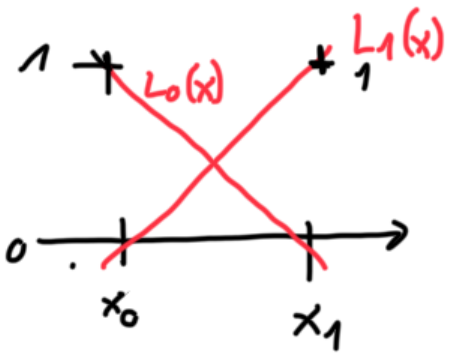
$$f'(x_j) = \underbrace{\sum_{k=0}^{n} f(x_k) L_k'(x_j)}_{\text{formula we can evaluate}} + \frac{f^{(n+1)}(\xi(x_j))}{(n+1)!} \prod_{\substack{k=0 \\ k \neq j}}^{n} (x_j - x_k)$$

evaluate error estimate

Examples:

$n=1$, x_0, $x_1 = x_0 + h$ $\quad h>0$

then $L_0(x) = \frac{x-x_1}{x_0-x_1}$ $\qquad L_1(x) = \frac{x-x_0}{x_1-x_0}$



$$L_0'(x) = \frac{1}{x_0-x_1}, \quad L_1(x) = \frac{1}{x_1-x_0} = \frac{1}{h}$$

$$= -\frac{1}{h}$$

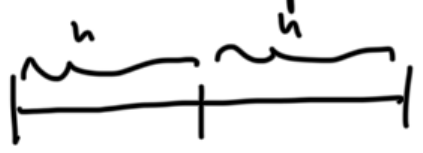
$$f'(x_0) = -\frac{f(x_0)}{h} + \frac{f(x_1)}{h} + \frac{f''(\xi)}{2}\underbrace{(x_0-x_1)}_{-h}$$

$$= \underbrace{\frac{f(x_1)-f(x_0)}{h}}_{\text{forward difference } (h>0)} \underbrace{- \frac{h}{2} f''(\xi)}_{\text{error}}$$

Error is $O(h)$ for $n=1$

$$f'(x_1) = \underbrace{\frac{f(x_1)-f(x_0)}{h}}_{\text{backward difference } h>0} + \underbrace{\frac{f''(\xi)}{2}\underbrace{(x_1-x_0)}_{h}}_{\text{error}}$$

$n=2$: results in 3 point formulas



$x_1 = x_0 + h$

$x_2 = x_0 + 2h$

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} \qquad L_0'(x) = \frac{2x-x_1-x_2}{(x_0-x_1)(x_0-x_2)}$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} \qquad L_1'(x) = \frac{2x-x_0-x_2}{(x_1-x_0)(x_1-x_2)}$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} \qquad L_2'(x) = \frac{2x-x_0-x_1}{(x_2-x_0)(x_2-x_1)}$$

Plug in x_0 (use $x_1 = x_0+h$, $x_2 = x_0+2h$)

$$L_0'(x_0) = \frac{2x_0-(x_0+h)-(x_0+2h)}{-h(-2h)} = \frac{-3h}{2h^2} = -\frac{3}{2h}$$

$$L_1'(x_0) = \frac{2x_0-x_0-(x_0+2h)}{h(-h)} = \frac{-2h}{-h^2} = \frac{2}{h}$$

$$L_2'(x_0) = \frac{2x_0-x_0-(x_0+h)}{2h\,h} = \frac{-h}{2h^2} = -\frac{1}{2h}$$

Then $f'(x_0) = f(x_0)\left(-\frac{3}{2h}\right) + f(x_1)\frac{2}{h} + f(x_2)\cdot\left(-\frac{1}{2h}\right) + \frac{h^2}{3}f^{(3)}(\xi)$

$$= \frac{-3f(x_0)+4f(x_1)-f(x_2)}{2h} + \frac{h^2}{3}f^{(3)}(\xi)$$

left endpoint

Plug in x_1 (midpoint)

$$L_0'(x_1) = -\frac{1}{2h} \qquad L_1'(x_1) = 0 \;, \quad L_2'(x_1) = \frac{1}{2h}$$

$$\Rightarrow f'(x_1) = \frac{f(x_2)-f(x_0)}{2h} - \frac{h^2}{6}f^{(3)}(\xi)$$

midpoint

Error is $O(h^2)$

etc.